

Attack–Fault Ambiguity Against a Strategic Attacker: The Stealth–Harm Frontier, Exploitable Commitments, and the Verification Margin

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Abstract—Analyses of attack–fault ambiguity at electronic trading gateways have treated the attacker as fixed. We study an attacker who knows the operator’s committed rule, as a Stackelberg game in a linear-Gaussian order-flow model with the operator as leader and an attacker staying within total variation ε of a fault’s observation law. We prove bounds valid for every strategy pair, hence under simultaneous play: the excess harm of an ε -stealthy attacker is bounded by a computable stealth–harm frontier that, under generic state-level injection, falls to zero as $\varepsilon \rightarrow 0$; no rule without verification detects it with probability above its false-alarm rate plus ε ; and verification on a separate channel adds a margin proportional to how often fault posteriors reach the verification interval. A threshold rule committed to a nominal attack model detects payloads its statistic does not see only at its false-alarm rate. When the stealth constraint binds, the threshold rule planning against the frontier’s maximizer, not the harm direction, alone leaves no harmful payload unseen and minimizes the attacker’s value at a given false-alarm rate; when the attacker’s capture loss equals the operator’s containment saving, a Stackelberg-optimal threshold commitment can be taken of this form where a minimizing cutoff exists. Under exact imitation, the game reduces to the non-strategic confounded case: the operator verifies at the prior or not at all. Numerically, the optimal threshold commitment usually ignores the observation, and the strategic verification region widens, but not preferentially where fault posteriors rarely reach the non-strategic interval.

Index Terms—Stackelberg games, stealthy attacks, fault diagnosis, cyber-physical systems security, attack detection, electronic trading.

I. INTRODUCTION

An incident at an electronic trading gateway, such as an algorithm that suddenly sends abnormal order flow, can have two causes: a *compromise*, in which an adversary controls the order flow, or a benign *fault*, such as a malfunctioning algorithm. The operator must decide whether to contain the source, at the cost of disrupting a legitimate client if the cause is a fault, or to leave it, at the cost of the harm a compromise goes on to do. Throughout, *the framework* means the setting of Papers 1 and 2 of this series: an operator observes the order flow over a window, forms the posterior probability that the cause is a

compromise, and chooses between containing and leaving, possibly after a costly verification. A *rule* is the operator’s committed decision rule in this framework: a map from what it observes (the order flow and, when it verifies, the outcome of the check) to the probability of containing (Definition 5).

Paper 1 [1], a working paper, models the order flow as a linear-Gaussian system and characterizes when a compromise and a fault cannot be told apart from order flow alone, and what that ambiguity costs the operator. Paper 2 [2], also a working paper, adds verification as a costed action that reads the authorization channel rather than the order flow, and derives a leave / verify / contain rule with two belief thresholds. Both treat the attacker as fixed: its injection is a given alternative to the fault, and the rule is designed against it. Paper 3 [3], in preparation, is to make the response sequential with graded containment; this paper uses none of its results.

This paper asks whether the framework survives an adversary who knows the rule. We model the interaction as a leader–follower game, as in Stackelberg security games [4]: the operator commits to a rule, and the attacker, knowing it, chooses its injection. Commitment is the natural reading at a trading gateway, where pre-trade controls and kill-switch procedures are documented, tested and audited before any incident; and a rule that is safe only while it is secret offers no security guarantee. The attacker is ε -stealthy: the law of the order flow under its injection lies within total-variation distance ε of the law under the fault it imitates.

We address four questions. First, how much excess harm, beyond what the imitated fault causes, can an ε -stealthy attacker with a given payload budget extract, and how does this stealth–harm frontier depend on ε (Section IV)? Second, what remains of the frontier as ε tends to zero, under output-level and under state-level injection? Third, can an attacker exploit a threshold rule committed to a nominal attack model, and what should the operator commit to instead? Fourth, what detection margin does a verification step, which reads a channel order flow cannot disguise, guarantee against a strategic attacker, and where should the operator verify?

The first two questions, the first half of the third and the first half of the fourth are answered by results that hold for every rule: the frontier and its limit as ε tends to zero (Proposition 2, Theorem 1), a detection cap for rules

without verification (Proposition 1), the exploitability of a threshold rule committed to a nominal attack model (Proposition 3), and the detection margin of verification (Proposition 4).

We give two further results. The first addresses the second half of the third question when the harm direction has no component in the blind subspace and the stealth constraint binds (Proposition 5). Among threshold rules at a given false-alarm rate, the rule that plans against the stealth-harm frontier’s maximizer M_W^+g , rather than against the harm direction g , minimizes the attacker’s value and is the only one that leaves no payload with positive excess harm unseen; when the attacker’s capture loss equals the operator’s containment saving, a Stackelberg-optimal commitment among threshold rules can be taken of this form, where a minimizing cutoff exists. Unlike Stackelberg prediction and strategic classification, which fix the cost of manipulation and optimize a classifier or its threshold [5], [6], the operator here chooses the direction of its statistic against a follower who trades harm against detection inside a stealth ball measured in the statistic’s own metric. The second is the limiting case $\varepsilon = 0$ of the second half of the fourth question, in which the game reduces to Paper 2’s confounded case: order flow carries no information, and the operator verifies at the prior or not at all (Proposition 6). Outside these two regimes, what the operator should commit to and where it should verify are computed in Section V, where the optimal threshold commitment usually ignores the observation; for every ε , Proposition 4 guarantees a verification margin wherever the interval is placed, and Proposition 6(e) bounds the operator’s Stackelberg cost from above.

Section II positions the paper, Section III states the game, Section IV gives the results (proofs in the Supplementary Material), Section V is the numerical study, Section VI discusses simultaneous play and the limits of the model, and Section VII concludes.

II. RELATED WORK

a) Stealthy attacks and defence design: Attacks that an output-based monitor cannot detect correspond to the zero dynamics of a linear system and, in the attack-free case, to its unobservable subspace [7]. Paper 1 [1] uses this subspace to characterize when a compromise and a fault remain confounded under state-level injection; this paper takes the mechanism as given and studies its consequence for a strategic attacker. Stealth with a tolerance was introduced as an ε -bound on the Kullback–Leibler divergence between attacked and nominal measurement sequences, with the largest performance degradation given as a function of ε [8]. Optimal ε -stealthy attacks have since been derived on a linear-quadratic-Gaussian cost, with an upper bound on the attack performance as a function of the stealth level [9], analytically for remote state estimation [10], and under a parameterized notion of ε -stealthiness [11]. On the defender’s side, a detector limits the biases a stealthy attacker can exert, which bounds how long a system can remain under stealthy attack

before its safety is violated [12]; the expected impact of stealthy injection has been used to allocate monitors in a Stackelberg framework [13] and sensors in networks [14]; and in a moving-target defence the defender randomizes its detector threshold in a Bayesian game [15]. Here the stealth budget is a total-variation distance, which bounds the detection gap of any rule directly; the reference law is that of a benign fault rather than of the attack-free system; the operator commits to a posterior containment rule rather than allocating sensors or randomizing a threshold; and the limit of the excess harm as ε tends to zero is tied to the pair (A, C) .

b) Adversarial classification and hypothesis testing:

Adversarial classification casts a classifier and an adversary who modifies its inputs as players of a game [16]: a naive-Bayes classifier anticipates one round of a cost-sensitive adversary who modifies discrete features to be classified benign. Stackelberg prediction games let the learner commit to a predictor while the data generator transforms the data distribution at a regularized cost to maximize its own loss, and compute the equilibrium by bilevel optimization [5]. In strategic classification, a classifier is committed and agents move their features at a cost to obtain the positive label, and the optimal classifier is a shifted threshold of the score [6]. The step these works share with Proposition 5, the worst-case shift of a committed linear score over an attacker’s ellipsoid, is standard in robust linear classification and is not claimed here. Proposition 5 differs in three ways. The follower maximizes excess harm net of detection, not the learner’s loss or the positive label, and is caught with a probability rather than by crossing a boundary. Its feasible set is a hard total-variation stealth constraint to a benign fault’s law, in the same metric M_W that defines the leader’s likelihood-ratio statistic. And the leader’s choice is the direction of its statistic, not its threshold: at a fixed cutoff every direction has the same false-alarm rate, the optimal direction has the closed form M_W^+g , and it is the only direction that leaves no harmful payload unseen; the shifted threshold of strategic classification corresponds to the cutoff, which Proposition 5(b) leaves to a one-dimensional minimization. Closest to this paper, game-theoretic Neyman–Pearson detection designs an evasion-aware detector against a strategic evader and summarizes what each side can achieve by an equilibrium receiver operating characteristic [17]. Over controlled continuous-time dynamics, a zero-sum game between a covert attacker and a detector has bias injection and a likelihood-ratio test as its unique saddle point, with type-II error exponents in the data length [18]. Here the game is leader–follower rather than zero-sum, the operator minimizes an expected cost with a posterior containment rule instead of a Neyman–Pearson criterion, the benign alternative is a fault, the observations come from a linear dynamical system over a finite window, and a verification channel is available.

c) Attacks, faults and active verification: A designed watermark combined with adaptive estimation discriminates sensor replay attacks from sensor bias faults, and

conditions characterize which attacks and faults can be told apart [19]. When to watermark can itself be posed as a stochastic control problem, whose optimal policy consists of two thresholds on the posterior probability of attack [20]; the two-threshold shape of a verification rule is therefore not new. In these works the attacker does not respond to the check; here the attacker knows the committed two-threshold rule with verification of Paper 2 [2] and best-responds to it, and the question is which detection margin remains.

d) Signalling, cheap talk and persuasion: When a sender's message carries no information, the receiver acts on its prior: this is the babbling equilibrium of cheap talk [21], the pooling equilibria of signalling games, in which types that send the same signal cannot be separated [22], and the uninformative signal of Bayesian persuasion [23]. An attacker that imitates the fault exactly makes order flow such a signal. Proposition 6 is this classical case in the present game, with verification as the only separating device, and it is presented as a limiting case that ties the game to Paper 2 [2], not as a new equilibrium concept.

e) Security games: In Stackelberg security games a defender commits to a strategy that the attacker observes before responding, and the relation between Stackelberg and Nash strategies in such games has been studied in detail [4]. This paper adopts the commitment form with the operator as leader and returns to simultaneous play in Section VI.

f) Manipulation in electronic markets: In high-frequency trading, the optimal spoofing strategy under multilevel order-book imbalance has been used to characterize and detect spoofing [24]. There the detector is not a committed leader, and no benign fault is confounded with the manipulation.

III. MODEL

This section states the game. The operator, as leader, commits to a rule and an observation window; the attacker, as follower, observes both and chooses an injection within a payload budget; nature then draws the cause, compromise or fault (Definition 1). Observations follow the linear-Gaussian order-flow model of Paper 1 [1] on a finite window W of observation times after the incident's onset, under output-level or state-level injection (Definition 2). The attacker's payload is the part of its injection that differs from the fault's (Definition 3), and it is ε -stealthy when the law of the observations on W under its injection lies within total-variation distance ε of the law under the fault (Definition 4). The operator chooses among rules without verification, including threshold rules committed to a nominal attack model, and rules with a verification step of the form proposed in Paper 2 [2] (Definition 5). The attacker's *excess harm* is a linear functional of its payload, the harm it extracts beyond what the imitated fault would cause (Definition 7); the stealth-harm frontier $\mathcal{F}_W(\varepsilon)$ is the largest excess harm over ε -stealthy payloads, and the harm ceiling is its limit as ε tends to zero (Definition 9).

A. The game

Definition 1 (Players, timing, game form). There are two players, the *operator* (leader) and the *attacker* (follower), and a move of nature. (1) The operator commits to a rule π from a class Π (Definition 5) and to an observation window W . (2) The attacker observes π and W and chooses an injection vector x_K from its admissible set (Definition 3). (3) Nature draws the cause $M = K$ with probability $p \in (0, 1)$ and $M = F$ otherwise; under F the fault instance x_F is fixed (Assumption 4), under K the injection is x_K . (4) The operator observes Y_W (and, under a rule with verification, the verification outcome Z) and applies π . The attacker's choice is a pure strategy; the operator's rule may be randomized (values in $[0, 1]$). The operator does not revise π after observing Y_W beyond what π itself prescribes (commitment).

Definition 2 (Observation model and stealth geometry). The system follows Paper 1's output-level model (Paper 1, Definition D2) or its state-level model (Paper 1, Definition D9), with a finite window $W \supseteq \{\tau, \dots, \tau + H - 1\}$, $H \geq 1$. For an injection vector $x \in \mathbb{R}^n$, write P_x^W for the law of Y_W when the injection from the onset on is x . Under Assumption 1 every P_x^W is Gaussian with a covariance $V_W \succ 0$ that does not depend on x , and mean $\mu_W + G_W x$, where G_W stacks, for $t \in W$, the zero block if $t < \tau$, and for $t = \tau + j$, $j \geq 0$,

$$C \text{ (output-level),} \quad C \sum_{i=0}^j A^i \text{ (state-level).}$$

The *stealth matrix* is $M_W = G_W^\top V_W^{-1} G_W \succeq 0$, and the *blind subspace* is $\mathcal{B}_W = \ker G_W = \ker M_W$.

Definition 3 (Attacker strategy, payload, budget). The attacker chooses a severity $\theta_K \in \mathbb{R}$ and a direction $b_K \in \mathbb{R}^n$; only the injection vector $x_K = \theta_K b_K$ enters the observation law (Definition 2), so a strategy is identified with x_K . The *payload* is $\Delta = x_K - x_F$, the part of the injection that differs from the fault's (Paper 1's Δ_0). The admissible set is the payload budget

$$\mathcal{X}_r = \{x_K \in \mathbb{R}^n : \|x_K - x_F\|_2 \leq r\}, \quad r > 0.$$

Definition 4 (ε -stealth). For $\varepsilon \in [0, 1]$, an attacker strategy x_K is ε -stealthy on W if $\text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon$ (total variation as in Paper 1, Definition D8). The ε -stealthy set is $\mathcal{E}_W(\varepsilon) = \{x_K \in \mathcal{X}_r : \text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon\}$. Under Assumption 1, with $d_W(\Delta) = \sqrt{\Delta^\top M_W \Delta}$,

$$\text{TV}(P_{x_K}^W, P_{x_F}^W) = 2\Phi(d_W(\Delta)/2) - 1,$$

$$\mathcal{E}_W(\varepsilon) = \{x_K \in \mathcal{X}_r : d_W(x_K - x_F) \leq \rho_\varepsilon\},$$

where

$$\rho_\varepsilon = 2\Phi^{-1}\left(\frac{1+\varepsilon}{2}\right), \quad \rho_1 = +\infty.$$

$\varepsilon = 0$ is exact confounding on W (Paper 1, Definition D3).

Definition 5 (Operator rules). Write the operator's loss table as $L(\text{contain}, K) = C_K$, $L(\text{contain}, F) = C_F$, $L(\text{leave}, K) = H_K$, $L(\text{leave}, F) = H_F$ (Paper 1, Definition D6), and put $a = H_K - C_K$ and $b = C_F - H_F$ (both

positive under Assumption 5). A rule with verification uses a check on a separate channel (Assumption 6) with outcome $Z \in \{+, -\}$, *sensitivity* $s_v = P(Z = + | K)$, *specificity* $t_v = P(Z = - | F)$, cost c_v , delay τ_v , and harm rate r_K of a true compromise during the delay (Paper 2, Definition 2). A *rule* on W is a measurable map π from observations to the probability of containing (Paper 1, Definition D7), possibly after a verification step. The classes used are: (i) *Committed threshold rule* $\pi_{\bar{x}}^{\text{thr}}$. The operator fixes a *nominal compromise model* $\bar{x} \neq x_F$ (the compromise instance it plans against) and contains iff the nominal posterior $\bar{q}(Y_W) = p f_{\bar{x}}^W(Y_W) / (p f_{\bar{x}}^W(Y_W) + (1 - p) f_{x_F}^W(Y_W))$ satisfies $\bar{q}(Y_W) \geq q^* = b/(a + b)$ (Paper 2, Lemma 1). (ii) *Committed two-threshold rule with verification* $\pi_{\bar{x}}^{\text{ver}}$. With the same $\bar{q}$ and thresholds $q_1 \leq q_2$: leave if $\bar{q} \leq q_1$; if $q_1 < \bar{q} < q_2$, verify, then contain on $Z = +$ and leave on $Z = -$; contain if $\bar{q} \geq q_2$. Paper 2's thresholds (Paper 2, Theorem 1) are, when its verification region is non-empty,

$$q_1 = \frac{c_v + (1 - t_v)b}{s_v a + (1 - t_v)b - r_K \tau_v},$$

$$q_2 = \frac{t_v b - c_v}{t_v b + (1 - s_v)a + r_K \tau_v},$$

and $q_1 = q_2 = q^*$ otherwise; the results below use only $q_1 \leq q_2$. (iii) *General rule*: any π (for the operator's optimal commitment); rules with verification are written as the quadruple $(\delta, \pi_0, \pi_+, \pi_-)$ of the verify probability δ , the containment probability π_0 without verification and the containment probabilities $\pi_{\pm}$ after $Z = \pm$, each a measurable map from observations to $[0, 1]$, as in Paper 2, Definition 4.

Definition 6 (Detection, false alarm, verification mass). For a rule π let $\phi_{\pi}^M(y) \in [0, 1]$ be the probability of containing given $Y_W = y$ and cause M (for a rule without verification $\phi_{\pi}^K = \phi_{\pi}^F = \pi$; for $\pi_{\bar{x}}^{\text{ver}}$, $\phi_{\pi}^K = \mathbb{1}[\bar{q} \geq q_2] + s_v \mathbb{1}[q_1 < \bar{q} < q_2]$ and $\phi_{\pi}^F = \mathbb{1}[\bar{q} \geq q_2] + (1 - t_v) \mathbb{1}[q_1 < \bar{q} < q_2]$). The *detection probability* of strategy x_K is $\beta_{\pi}(x_K) = \mathbb{E}_{x_K}[\phi_{\pi}^K(Y_W)]$, the *false-alarm probability* is $\alpha_{\pi} = \mathbb{E}_{x_F}[\phi_{\pi}^F(Y_W)]$, and for a rule with verification the *fault verification mass* is $m_F = P_{x_F}(q_1 < \bar{q}(Y_W) < q_2)$ (Paper 2's verification mass, Definition 5, evaluated on the fault law).

B. Harm, payoffs and the stealth-harm frontier

Definition 7 (Harm and payoffs). Fix a *harm vector* $g \in \mathbb{R}^n$. The attacker's *excess harm* is the linear functional $h(\Delta) = g^T \Delta$ of its payload: the harm it extracts beyond what the fault it imitates would cause. The attacker's payoff against rule π is the expected excess harm extracted before containment, net of a capture loss $c_A \geq 0$:

$$U_A(\pi, x_K) = h(x_K - x_F) (1 - \beta_{\pi}(x_K)) - c_A \beta_{\pi}(x_K).$$

The operator's expected cost is Paper 1's loss table (Definition D6, general symbols) with the leave-a-compromise entry raised by the extracted harm, $H_K + h(\Delta)$:

$$J(\pi, x_K) = p \left(\beta_{\pi}(x_K) C_K + (1 - \beta_{\pi}(x_K)) (H_K + h(\Delta)) \right) + (1 - p) \left(\beta_{\pi}(x_K) C_F + (1 - \beta_{\pi}(x_K)) (H_F + h(\Delta)) \right)$$

where Ver_{π} is the expected verification cost $c_v + \mathbb{1}[M = K] r_K \tau_v$ incurred on the event that π verifies (Paper 2, Definition 4), and zero for a rule without verification. The game is not zero-sum ($U_A \neq -J$ in general).

Definition 8 (Strong Stackelberg equilibrium). For a rule π , the attacker's ε -stealthy best-response set is $\text{BR}_{\varepsilon}(\pi) = \arg \max_{x \in \mathcal{E}_W(\varepsilon)} U_A(\pi, x)$ (non-empty under Assumption 3 when $U_A(\pi, \cdot)$ is upper semicontinuous). A pair (π^*, x^*) is a *strong Stackelberg equilibrium* in class Π at stealth level ε if $x^* \in \text{BR}_{\varepsilon}(\pi^*)$, ties being broken in the operator's favour (x^* minimizes $J(\pi^*, \cdot)$ over $\text{BR}_{\varepsilon}(\pi^*)$), and π^* minimizes $\min_{x \in \text{BR}_{\varepsilon}(\pi)} J(\pi, x)$ over $\pi \in \Pi$ (an infimum if not attained). Under simultaneous play, the corresponding notion is a Nash equilibrium of the same payoffs, the attacker possibly randomizing over $\mathcal{E}_W(\varepsilon)$.

The bounds of Section IV hold for every strategy pair and do not depend on J . Propositions 5 and 6 characterize strong Stackelberg commitments in two regimes, and Proposition 7 compares them with simultaneous play.

Definition 9 (Stealth-harm frontier, detection-harm frontier, harm ceiling). The *stealth-harm frontier* on W is

$$\mathcal{F}_W(\varepsilon) = \sup \{ h(x - x_F) : x \in \mathcal{E}_W(\varepsilon) \}, \quad \varepsilon \in [0, 1],$$

which does not depend on the operator's rule. The *detection-harm frontier against a rule* π is $\Psi_{\pi}(\beta) = \sup \{ h(x - x_F) : x \in \mathcal{X}_{\pi}, \beta_{\pi}(x) \leq \beta \}$, $\beta \in [0, 1]$. The *harm ceiling* is $\bar{\mathcal{F}}_W = \lim_{\varepsilon \downarrow 0} \mathcal{F}_W(\varepsilon)$ (the limit exists since $\mathcal{F}_W$ is nondecreasing).

C. Standing assumptions

Assumption 1 (Paper 1's linear-Gaussian model). Paper 1's standing assumptions (A1)–(A6) hold; written out: (A1) the process noise $w_t \sim \mathcal{N}(0, Q)$ and the observation noise $\varepsilon_t \sim \mathcal{N}(0, \Sigma)$, Q, Σ symmetric positive semidefinite, are mutually independent across time and across the two sequences, with a law that does not depend on the cause or the injection; (A2) the initial state is $x_0 = 0$ (any Gaussian x_0 independent of the noise, with a law that does not depend on the cause or the injection, would do); (A3) A, Q, C, Σ are identical under both causes; only the injection vector depends on the cause; (A4) the onset τ is deterministic and common to both causes; (A5) the window reaches the onset, here in the stronger form used by Paper 1's Theorem 3: W is finite and contains $\{\tau, \dots, \tau + H - 1\}$, $H \geq 1$ (Definition 2); (A6) $\Sigma \succ 0$, hence $V_W \succ 0$. The quantities (A, C, Q, Σ) , τ and W are known to both players (Assumption 2).

Assumption 2 (Information). (1) *Common knowledge*. The following are common knowledge between the operator and the attacker: the model of Assumption 1 $((A, C, Q, \Sigma), \tau, W)$; the fault instance x_F ; the prior p ; the loss table (C_K, C_F, H_K, H_F) ; the verification parameters $(s_v, t_v, c_v, \tau_v, r_K)$; the harm vector g ; the payload budget b ; the capture loss c_A ; the operator's stealth level ε . The game

is therefore one of complete information: the attacker's payoff U_A , its admissible set $\mathcal{E}_W(\varepsilon)$ and hence its best-response set $\text{BR}_\varepsilon(\pi)$ (Definition 8) are known to the operator for every rule π . (2) *Commitment*. The operator chooses its rule π (including the nominal model $\bar{x}$ and, for a rule with verification, the thresholds) before the attacker moves, and the attacker observes π as a map from observations and verification outcomes to actions before choosing x_K . (3) *What is not observed*. Neither player observes the cause M drawn by nature. The operator does not observe the attacker's realized choice x_K ; its only information at decision time is Y_W and, if it verifies, Z . In a strong Stackelberg equilibrium the operator *anticipates* the attacker's choice by computing $\text{BR}_\varepsilon(\pi)$ from the common-knowledge primitives in (1), and chooses π accordingly (Definition 8); what it cannot do is condition its action on x_K other than through Y_W and Z . In particular a rule may be committed to a nominal model $\bar{x}$ different from the anticipated best response. (4) *What the attacker's best response uses*. The attacker's best response depends on p , the loss table, $\bar{x}$ and the thresholds only through π ; for a rule with verification it depends on the verification parameters through s_v (in ϕ_π^K , Definition 6); and it depends on g , r , c_A , ε and x_F directly.

Assumption 3 (Payload budget). The attacker's admissible set is $\mathcal{X}_r$ (Definition 3) for an $r > 0$ known to both players (Assumption 2). (Compact and convex; any compact convex admissible set containing x_F could replace it at the cost of the closed forms.)

Assumption 4 (One fixed fault instance). Under F the injection is a fixed, known $x_F = \theta_F b_F$ (Paper 1's pairwise-instance reading, Definition D1); no distribution over fault instances is modelled.

Assumption 5 (Costs and verification). (1) *Cost orderings* (Paper 2, Assumption 1): $a = H_K - C_K > 0$ and $b = C_F - H_F > 0$: leaving a compromise is strictly worse than containing it, and containing a fault is strictly worse than leaving it. For rules with verification, in addition: (2) *Informative one-shot verification* (Paper 2, Assumption 3): $s_v, t_v \in [0, 1]$ with $s_v + t_v > 1$; $c_v \geq 0$, $\tau_v \geq 0$, $r_K \geq 0$; verification is available once, and after its outcome the operator takes a terminal action (contain or leave). (3) *Delay accounting* (Paper 2, Assumption 4): the harm $r_K \tau_v$ accrued by a true compromise during the verification delay is additional to the terminal cost, whichever terminal action is taken; the terminal costs and (s_v, t_v) do not depend on τ_v ; a fault accrues no additional cost during the delay; and no further order-flow evidence is used between the start of verification and the terminal action.

Assumption 6 (Verification is not manipulable by order flow). Paper 2's separate-channel assumption (Assumption 2) holds for every attacker strategy: verification reads authorization, not order flow, through a channel the party submitting the orders does not control; given the cause, Z is independent of Y_W , $P(Z, Y_W | M) = P(Z | M) P(Y_W | M)$ for $M \in \{K, F\}$; and (s_v, t_v) depend neither on Y_W

nor on x_K . (An attacker who also attacks the authorization channel, for instance with stolen credentials, violates this assumption; that case is out of scope.)

IV. RESULTS

This section answers the questions of Section I in the order stated there (Sections IV-A–IV-C), then gives the aligned commitment (Section IV-D) and the limiting case of exact imitation (Section IV-E); the optimal commitment outside these regimes is computed in Section V. The proofs of all results are given in the Supplementary Material.

A. The stealth-harm frontier and the harm ceiling

Proposition 1 bounds, for every committed rule, the detection probability of an attacker whose observation law is within total variation ε of the fault's.

Proposition 1 (Total-variation detection cap). *Assume Assumption 1. Let $\varepsilon \in [0, 1]$ and let $x_K \in \mathbb{R}^n$ satisfy $\text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon$ (in particular, any $x_K \in \mathcal{E}_W(\varepsilon)$). (a) For every measurable ϕ from observations on W to $[0, 1]$, $|\mathbb{E}_{x_K}[\phi(Y_W)] - \mathbb{E}_{x_F}[\phi(Y_W)]| \leq \varepsilon$. (b) For every rule π without verification, $\alpha_\pi - \varepsilon \leq \beta_\pi(x_K) \leq \alpha_\pi + \varepsilon$. (c) Assume in addition Assumption 6. For every rule with verification, written as in Definition 5(iii) as $(\delta, \pi_0, \pi_+, \pi_-)$, let $\phi^K = (1 - \delta)\pi_0 + \delta(s_v\pi_+ + (1 - s_v)\pi_-)$ and $\phi^F = (1 - \delta)\pi_0 + \delta((1 - t_v)\pi_+ + t_v\pi_-)$ (for π_x^{ver} these are the maps of Definition 6). Then $\beta_\pi(x_K) = \mathbb{E}_{x_K}[\phi^K(Y_W)]$, $\alpha_\pi = \mathbb{E}_{x_F}[\phi^F(Y_W)]$, and $\mathbb{E}_{x_F}[\phi^K(Y_W)] - \varepsilon \leq \beta_\pi(x_K) \leq \mathbb{E}_{x_F}[\phi^K(Y_W)] + \varepsilon$. (d) (Tightness.) For every $x \in \mathbb{R}^n$, $\sup_\pi (\beta_\pi(x) - \alpha_\pi) = \text{TV}(P_x^W, P_{x_F}^W)$, the supremum over rules without verification, attained by the rule that contains iff $f_x^W(Y_W) > f_{x_F}^W(Y_W)$.*

Proof: Supplementary Material, Section S1.1.

Proposition 2 gives the shape of the stealth-harm frontier $\mathcal{F}_W(\varepsilon)$ of Definition 9 as a function of ε .

Lemma 1 gives the largest excess harm over payloads that leave the observation law unchanged, which Proposition 2 and Theorem 1 use.

Lemma 1 (Excess harm on the blind subspace). *Assume Assumption 1 and Assumption 3, and write $\Pi_{\mathcal{B}_W}$ for the orthogonal projection onto the blind subspace $\mathcal{B}_W = \ker G_W$. Then $\{\Delta \in \mathbb{R}^n : \|\Delta\|_2 \leq r, \Delta^\top M_W \Delta = 0\} = \{\Delta \in \mathcal{B}_W : \|\Delta\|_2 \leq r\}$, and*

$$\begin{aligned} \mathcal{F}_W(0) &= \max\{g^\top \Delta : \|\Delta\|_2 \leq r, \Delta^\top M_W \Delta = 0\} \\ &= r \|\Pi_{\mathcal{B}_W} g\|_2, \end{aligned}$$

attained at $\Delta = r \Pi_{\mathcal{B}_W} g / \|\Pi_{\mathcal{B}_W} g\|_2$ when $\Pi_{\mathcal{B}_W} g \neq 0$ and at $\Delta = 0$ otherwise.

Proof: Supplementary Material, Section S1.2.

Proposition 2 (Shape of the stealth-harm frontier). *Assume Assumption 1 and Assumption 3, write $M = M_W$ and, for $\rho \in [0, \infty)$,*

$$f(\rho) = \max\{g^\top \Delta : \|\Delta\|_2 \leq r, \Delta^\top M \Delta \leq \rho^2\},$$

so that $\mathcal{F}_W(\varepsilon) = f(\rho_\varepsilon)$ for $\varepsilon \in [0, 1]$ (the maximum is attained). Let $\rho_{\text{sat}} = r\sqrt{g^\top M g}/\|g\|_2$ ($\rho_{\text{sat}} = 0$ if $g = 0$) and $\varepsilon_{\text{sat}} = 2\Phi(\rho_{\text{sat}}/2) - 1 \in [0, 1]$. (a) f is nondecreasing, concave and continuous on $[0, \infty)$. $\mathcal{F}_W$ is nondecreasing and continuous on $[0, 1]$, with $\mathcal{F}_W(1) = r\|g\|_2$. (b) (Saturation.) $\mathcal{F}_W(\varepsilon) = r\|g\|_2$ if $\varepsilon \geq \varepsilon_{\text{sat}}$ and $\mathcal{F}_W(\varepsilon) < r\|g\|_2$ if $\varepsilon < \varepsilon_{\text{sat}}$. Moreover $\varepsilon_{\text{sat}} = 0$ if and only if $g \in \ker G_W$. (c) (Without the budget.) For $\rho \geq 0$, $\sup\{g^\top \Delta : \Delta^\top M \Delta \leq \rho^2\} = \rho\sqrt{g^\top M^+ g}$ if $g \in \text{range}(M)$, and $+\infty$ otherwise; for $g \in \text{range}(M) \setminus \{0\}$ and $\rho > 0$ it is attained at $\rho M^+ g / \sqrt{g^\top M^+ g}$. (d) (With the budget: a one-dimensional search.) Let $g \neq 0$. Put $\rho_b = r\sqrt{g^\top M^+ g} / \|M^+ g\|_2$ if $g \in \text{range}(M)$ and $\rho_b = 0$ otherwise; for $\nu \geq 0$ put $\Delta(\nu) = (I + \nu M)^{-1} g$ and $R(\nu) = \Delta(\nu)^\top M \Delta(\nu) / \|\Delta(\nu)\|_2^2$. Then R is continuous and non-increasing on $[0, \infty)$, $R(0) = (\rho_{\text{sat}}/r)^2$, $\lim_{\nu \rightarrow \infty} R(\nu) = (\rho_b/r)^2$, so $\rho_b \leq \rho_{\text{sat}}$, and R is strictly decreasing if $\rho_b < \rho_{\text{sat}}$. Further: (d1) if $\rho \geq \rho_{\text{sat}}$, $f(\rho) = r\|g\|_2$, attained at $rg/\|g\|_2$; (d2) if $g \in \text{range}(M)$ and $\rho \leq \rho_b$, $f(\rho) = \rho\sqrt{g^\top M^+ g}$, attained as in (c) (budget inactive); (d3) if $\rho_b < \rho < \rho_{\text{sat}}$, the equation $R(\nu) = (\rho/r)^2$ has a unique root $\nu_\rho \in (0, \infty)$, and $f(\rho) = r g^\top \Delta(\nu_\rho) / \|\Delta(\nu_\rho)\|_2$, attained at $r \Delta(\nu_\rho) / \|\Delta(\nu_\rho)\|_2$ (both constraints active). Together with $f(0) = r\|\Pi_{\ker G_W} g\|_2$ (Lemma 1), the cases (d1)–(d3) cover every $\rho \geq 0$.

Proof: Supplementary Material, Section S1.3.

Theorem 1 gives the harm ceiling $\bar{\mathcal{F}}_W$ under output-level and state-level injection. Parts (b) and (c) are the analogues, for the excess harm of an attacker, of Paper 1, Theorem 1 (output level) and Paper 1, Theorem 3 (the finite-horizon version of its Theorem 2) and Corollary 3 (state level) [1]; the proof re-derives the linear algebra it needs and does not rely on those results. Part (d) is the analogue of the remark that follows Paper 1, Corollary 3, where “generic” means Lebesgue-almost every A for a fixed $C \neq 0$, as defined there; as Paper 1 stresses, this is a statement about a full-measure set of matrices, not a claim that the dynamics of a deployed market are generic.

Theorem 1 (Harm ceiling). *Assume Assumption 1 and Assumption 3, and write $\Pi_{\mathcal{B}_W}$ for the orthogonal projection onto the blind subspace $\mathcal{B}_W = \ker G_W$. (a) $\bar{\mathcal{F}}_W = \lim_{\varepsilon \downarrow 0} \mathcal{F}_W(\varepsilon) = \mathcal{F}_W(0) = r\|\Pi_{\mathcal{B}_W} g\|_2$, attained at $\Delta = r\Pi_{\mathcal{B}_W} g / \|\Pi_{\mathcal{B}_W} g\|_2$ when $\Pi_{\mathcal{B}_W} g \neq 0$ (the value is Lemma 1). (b) (Output level, Paper 1 Definition D2.) $\mathcal{B}_W = \ker C$, so $\bar{\mathcal{F}}_W = r\|\Pi_{\ker C} g\|_2$. (c) (State level, Paper 1 Definition D9.) If $W \supseteq \{\tau, \dots, \tau + H - 1\}$, then $\mathcal{B}_W \subseteq \bigcap_{j=0}^{H-1} \ker(CA^j)$, with equality if W contains no time $t \geq \tau + H$. If $H \geq n$, then $\mathcal{B}_W = \mathcal{N}(A, C)$, the unobservable subspace (Paper 1, Definition D10), so $\bar{\mathcal{F}}_W = r\|\Pi_{\mathcal{N}(A, C)} g\|_2$, which is zero whenever (A, C) is observable. (d) (Genericity.) For fixed $C \neq 0$, (A, C) is observable for Lebesgue-almost every $A \in \mathbb{R}^{n \times n}$. Hence, at the state level with $H \geq n$, the harm ceiling is zero excess harm for Lebesgue-almost every A . (e) (Absolute harm.) For every $\varepsilon \in [0, 1]$, $\sup\{g^\top x : x \in \mathcal{E}_W(\varepsilon)\} = g^\top x_F + \mathcal{F}_W(\varepsilon)$; so under the absolute-harm payoff $g^\top x_K$*

in place of h the ceiling is $g^\top x_F + \bar{\mathcal{F}}_W$, which equals the fault’s own harm $g^\top x_F$ whenever $\bar{\mathcal{F}}_W = 0$ (in particular at the state level with $H \geq n$ and (A, C) observable, the zero-ceiling case of (c), hence for Lebesgue-almost every A by (d)).

Proof: Supplementary Material, Section S1.4.

Exact stealth ($\varepsilon = 0$) is therefore always available, since the attacker can reproduce the fault’s injection; under state-level injection on a window of at least n steps and for generic dynamics it is the only exactly stealthy attack, so it yields zero excess harm, and any positive excess harm costs a strictly positive ε .

B. Exploitable commitments

Proposition 3 shows that a threshold rule committed to a nominal attack model detects a payload only through one scalar, so that an attacker who knows the rule can choose a payload the rule detects only at its false-alarm rate.

Proposition 3 (The committed threshold rule is blind off its nominal signature). *Assume Assumption 1. Fix a nominal model $\bar{x}$, a threshold $q^\dagger \in (0, 1)$ ($q^\dagger = q^*$ for $\pi_{\bar{x}}^{\text{thr}}$) and the rule π that contains iff $\bar{q}(Y_W) \geq q^\dagger$. Put $w = M_W(\bar{x} - x_F)$, $\bar{D} = d_W(\bar{x} - x_F)$, $\xi^\dagger = \log((1-p)q^\dagger/(p(1-q^\dagger)))$, and $\Delta = x_K - x_F$. (a) If $\bar{D} > 0$, then for every $x_K \in \mathbb{R}^n$*

$$\beta_\pi(x_K) = \Phi\left(\frac{w^\top \Delta - \xi^\dagger}{\bar{D}} - \frac{\bar{D}}{2}\right), \quad \alpha_\pi = \Phi\left(-\frac{\xi^\dagger}{\bar{D}} - \frac{\bar{D}}{2}\right).$$

So β_π depends on x_K only through the scalar $w^\top \Delta = (G_W \bar{x} - G_W x_F)^\top V_W^{-1} G_W \Delta$ and is strictly increasing in it: $\beta_\pi(x_K) = \alpha_\pi$ iff $w^\top \Delta = 0$, and $\beta_\pi(x_K) < \alpha_\pi$ iff $w^\top \Delta < 0$. (b) If $\bar{D} = 0$, then $\bar{q}(Y_W) = p$ almost surely, π is constant, and $\beta_\pi(x_K) = \alpha_\pi$ for every x_K . (c) Under Assumption 3, the detection-harm frontier at the false-alarm rate is $\Psi_\pi(\alpha_\pi) = r\|g\|_2$ if $w^\top g \leq 0$, and $\Psi_\pi(\alpha_\pi) = r\|\Pi_{w^\perp} g\|_2$ if $w^\top g > 0$, where $\Pi_{w^\perp}$ is the orthogonal projection onto $\{u : w^\top u = 0\}$; in the second case the supremum is attained by a payload with $\beta_\pi = \alpha_\pi$. (d) Under Assumption 3, let $\lambda_\perp^2 = \max\{u^\top M_W u : \|u\|_2 \leq 1, w^\top u = 0\}$ (the maximum over the unit sphere of $w^\perp$ when $w^\perp \neq \{0\}$, and 0 when $n = 1$ and $w \neq 0$). The largest total variation among payloads the rule does not see is $\max\{\text{TV}(P_x^W, P_{x_F}^W) : x \in \mathcal{X}_r, \beta_\pi(x) = \alpha_\pi\} = 2\Phi(r\lambda_\perp/2) - 1$, and it is strictly positive whenever $\text{rank } M_W \geq 2$.

Proof: Supplementary Material, Section S1.5.

C. The verification margin

Proposition 4 bounds the detection probability of every ε -stealthy attacker against a committed two-threshold rule with verification. The rule is the one proposed in Paper 2 [2]; the inequality itself uses only the ordering of the two thresholds and Assumption 6.

Proposition 4 (Verification margin). *Assume Assumption 1 and Assumption 6, with $s_v, t_v \in [0, 1]$. Let $\pi = \pi_{\bar{x}}^{\text{ver}}$*

with thresholds $q_1 \leq q_2$ (Paper 2's, or any others), $\alpha = \alpha_\pi$ and m_F the fault verification mass (Definition 6). For every $\varepsilon \in [0, 1]$ and every x_K with $\text{TV}(P_{x_K}^W, P_{x_F}^W) \leq \varepsilon$ (in particular every $x_K \in \mathcal{E}_W(\varepsilon)$), with $\Delta = x_K - x_F$: (a) $(s_v + t_v - 1)m_F - \varepsilon \leq \beta_\pi(x_K) - \alpha \leq (s_v + t_v - 1)m_F + \varepsilon$; (b) $1 - \alpha - (s_v + t_v - 1)m_F + \varepsilon \geq 0$ and $U_A(\pi, x_K) \leq h(\Delta)^+ (1 - \alpha - (s_v + t_v - 1)m_F + \varepsilon)$, where $h^+ = \max\{h, 0\}$; in particular, if $h(\Delta) \geq 0$, $U_A(\pi, x_K) \leq h(\Delta)(1 - \alpha - (s_v + t_v - 1)m_F + \varepsilon)$. Under Assumption 5 ($s_v + t_v > 1$), the margin $(s_v + t_v - 1)m_F$ is positive iff $m_F > 0$; if the verification region is empty ($q_1 = q_2$), $m_F = 0$ and (a) reduces to Proposition 1(b).

Proof: Supplementary Material, Section S1.6.

The bound holds wherever the verification interval is placed, and it depends on the interval only through the false-alarm rate α and the fault verification mass m_F . It guarantees a margin, not a placement. Where the operator should place the interval against a strategic attacker is answered in the limiting case $\varepsilon = 0$ by Proposition 6, and computed for $\varepsilon > 0$ in Section V.

D. The aligned commitment

Proposition 5 says which rule the operator should commit to instead of a nominal one when the stealth constraint binds.

The operator's choice among threshold rules is the direction of its statistic, that is, its nominal model, and its cutoff, which fixes the false-alarm rate. Assume the harm direction has no component in the blind subspace ($g \in \text{range } M_W$, $g \neq 0$) and the stealth constraint binds ($r^2\lambda_+(M_W) \geq \rho_\varepsilon^2$). Then the threshold rule that plans against the stealth-harm frontier's maximizer, $\bar{x} - x_F \in cM_W^+g + \mathcal{B}_W$ ($c > 0$, the *aligned* rule), is the only threshold rule that leaves no payload with positive excess harm unseen, and it minimizes the attacker's value among threshold rules with the same cutoff, hence the same false-alarm rate. When the capture loss equals the operator's containment saving ($c_A = a$), a Stackelberg-optimal commitment among threshold rules can be taken aligned, where a minimizing cutoff exists (otherwise the infimum is approached only as the aligned rule tends to never or always containing). These hold both for threshold rules with a free threshold (the class of Proposition 3) and within Definition 5(i) (threshold q^*), comparing rules of the same scaling $\bar{D}$. The naive commitment to the harm direction g is aligned if and only if g is an eigenvector of M_W ; otherwise it leaves a harmful payload unseen. Proposition 5 states this precisely.

Proposition 5 (Aligned commitment when stealth binds). Assume Assumption 1 and Assumption 3. Let $L_W = V_W^{-1/2}G_W$, so that $M_W = L_W^\top L_W$ and $d_W(\Delta) = \|L_W\Delta\|_2$. Let $\varepsilon \in (0, 1)$ and assume that $g \in \text{range}(M_W) \setminus \{0\}$ and that the stealth-binding condition $r^2\lambda_+ \geq \rho_\varepsilon^2$ holds, where λ_+ is the smallest positive eigenvalue of M_W . Put $h = (L_W^+)^T g$ and $\hat{h} = h/\|h\|_2$. Write $\mathcal{T}$ for the class of threshold rules π of Proposition 3 with nominal model $\bar{x}$,

$\bar{D} = d_W(\bar{x} - x_F) > 0$, and any threshold $q^\dagger \in (0, 1)$, and $\mathcal{T}^* \subset \mathcal{T}$ for those with $q^\dagger = q^*$, which are the committed threshold rules $\pi_{\bar{x}}^{\text{thr}}$ of Definition 5(i). Parts (a)–(d) concern $\mathcal{T}$; part (e) restates them within $\mathcal{T}^*$. For $\pi \in \mathcal{T}$, write $v_\pi = L_W(\bar{x} - x_F)/\bar{D}$ for its whitened signature, a unit vector of range L_W , and $k_\pi = \xi^\dagger/\bar{D} + \bar{D}/2$ for its cutoff. Then $\beta_\pi(x_K) = \Phi(v_\pi^\top L_W\Delta - k_\pi)$ and $\alpha_\pi = \Phi(-k_\pi)$, and every pair (v, k) of a unit vector $v \in \text{range } L_W$ and a $k \in \mathbb{R}$ is the signature and cutoff of some rule of $\mathcal{T}$. Call π aligned if $v_\pi = \hat{h}$, equivalently if $\bar{x} - x_F \in cM_W^+g + \mathcal{B}_W$ for some $c > 0$. Write $V_A(\pi) = \max_{x \in \mathcal{E}_W(\varepsilon)} U_A(\pi, x)$ for the attacker's value. (a) (The aligned rule minimizes the attacker's value.) Let $c_A \geq 0$. For every $\pi \in \mathcal{T}$ and every aligned rule $\pi_{\text{al}} \in \mathcal{T}$ with the same cutoff, $V_A(\pi) \geq V_A(\pi_{\text{al}})$. (b) (Optimal commitment when capture loss equals containment saving.) If $c_A = a$, then $J(\pi, x) \geq J(\pi_{\text{al}}, x')$ for every π and π_{al} as in (a) and all $x \in \text{BR}_\varepsilon(\pi)$, $x' \in \text{BR}_\varepsilon(\pi_{\text{al}})$. Hence the infimum over $\mathcal{T}$ of the Stackelberg objective $\min_{x \in \text{BR}_\varepsilon(\pi)} J(\pi, x)$ equals $\inf_{k \in \mathbb{R}} J_{\text{al}}(k)$, where $J_{\text{al}}(k)$ is the (common) cost of the aligned rules with cutoff k , a continuous function of k . Where a minimizing cutoff exists, a strong Stackelberg commitment in $\mathcal{T}$ can be taken aligned, at that cutoff. Where none exists, no rule of $\mathcal{T}$ attains the infimum. Then every sequence $\pi_n \in \mathcal{T}$ whose objective tends to the infimum has $|k_{\pi_n}| \rightarrow \infty$, the aligned rules with the cutoffs k_{π_n} also approach the infimum, and along any subsequence with $k_{\pi_n} \rightarrow +\infty$ (resp. $-\infty$) both α_{π_n} and β_{π_n} tend to 0 (resp. 1) uniformly on $\mathcal{E}_W(\varepsilon)$: the rules tend to never containing or to always containing. (c) (Only the aligned rule sees every harmful payload.) $\sup\{h(x - x_F) : x \in \mathcal{E}_W(\varepsilon), \beta_\pi(x) \leq \alpha_\pi\}$ is 0 if π is aligned and strictly positive otherwise. (d) (When the naive commitment is aligned.) For $c > 0$, the rule committed to $\bar{x} = x_F + cg$ is aligned if and only if g is an eigenvector of M_W . When it is not aligned, it leaves a payload with positive excess harm unseen, by (c). (e) (Within Definition 5(i), matched scaling.) Assume also Assumption 5(1), so that $q^* = b/(a + b) \in (0, 1)$. Put $\xi^* = \log((1 - p)q^*/(p(1 - q^*)))$. For $\pi = \pi_{\bar{x}}^{\text{thr}} \in \mathcal{T}^*$ with $\bar{D} = d_W(\bar{x} - x_F) > 0$, let $\pi_{\text{al}} = \pi_{\bar{x}_{\text{al}}}^{\text{thr}}$ with $\bar{x}_{\text{al}} = x_F + (\bar{D}/\|h\|_2)M_W^+g$. Then $\pi_{\text{al}} \in \mathcal{T}^*$ is aligned, has the same scaling $\bar{D}$, the same cutoff $k = \xi^*/\bar{D} + \bar{D}/2$ and the same false-alarm rate as π . Hence (a) and (c) hold within $\mathcal{T}^*$, and (b) holds with $\mathcal{T}^*$ in place of $\mathcal{T}$ and the aligned rules of $\mathcal{T}^*$ indexed by $\bar{D} > 0$: where a minimizing $\bar{D}$ exists, a strong Stackelberg commitment in $\mathcal{T}^*$ can be taken aligned at that $\bar{D}$; where none exists, no rule of $\mathcal{T}^*$ attains the infimum. Then for every sequence $\pi_n \in \mathcal{T}^*$ whose objective tends to the infimum, the aligned rules of $\mathcal{T}^*$ with the same scalings $\bar{D}_n$ also approach it, and every subsequence of $(\bar{D}_n)$ has a further subsequence with $\bar{D}_n \rightarrow \infty$ or $\bar{D}_n \downarrow 0$. Along $\bar{D}_n \rightarrow \infty$ the cutoffs tend to $+\infty$; along $\bar{D}_n \downarrow 0$ they tend to $+\infty$ if $p < q^*$, to $-\infty$ if $p > q^*$, and to 0 if $p = q^*$. So, except when $p = q^*$, the infimum over $\mathcal{T}^*$ is approached only as the rules tend to never or always containing, as in (b).

Proof: Supplementary Material, Section S1.7.

The aligned direction M_W^+g is the maximizer of the stealth-harm frontier without budget (Proposition 2(c)): the operator plans against the payload the frontier says is most harmful per unit of stealth, measured in the metric M_W of its own likelihood-ratio statistic, not against the harm direction g itself. Part (b) is proved for $c_A = a$, where the operator's cost differs from the attacker's value only by terms that depend on the cutoff alone. For other capture losses, and where the payload budget binds, whether the aligned rule remains Stackelberg-optimal is not established here; the numerical study of Section V computes the optimal commitment without these restrictions.

E. The limiting case of exact imitation

At $\varepsilon = 0$ exact confounding reduces to Paper 2's confounded case. Against an attacker that imitates the fault exactly, order flow carries no information, and the operator's Stackelberg-optimal rule is Paper 2's leave/verify/contain action at the prior [2], for the loss table in which leaving a compromise also costs the harm ceiling $\bar{\mathcal{F}}_W$. It therefore verifies always or never. Its cost is Paper 2's confounded optimum at that raised cost, and it bounds the Stackelberg cost from above for every ε , up to the excess $\mathcal{F}_W(\varepsilon) - \bar{\mathcal{F}}_W$ of the frontier over the ceiling. The game-theoretic content of this case is classical: exact imitation is a pooling outcome, in which the two causes send the same signal and the receiver best-responds to the prior over the pool [22]; order flow is then a babbling message [21], and, as with an uninformative signal in Bayesian persuasion [23], the operator acts on the prior. Proposition 6 is therefore a consistency check that ties the game to Paper 2 and bounds the operator's cost from above, not a headline result. What it adds to the classical case is the loss table raised by the harm ceiling, the reduction to Paper 2's confounded optimum, and the bound of part (e) for $\varepsilon > 0$.

Proposition 6 (Limiting case $\varepsilon = 0$: exact imitation reduces to Paper 2's confounded case). *Assume Assumption 1, Assumption 3, Assumption 5 and Assumption 6. Write $\bar{\mathcal{F}}_W = r\|\Pi_{\mathcal{B}_W}g\|_2$ (Theorem 1(a)) and $a' = a + \bar{\mathcal{F}}_W$. For a rule $\pi = (\delta, \pi_0, \pi_+, \pi_-)$ of Definition 5(iii) (a rule without verification has $\delta \equiv 0$), write ϕ^K as in Proposition 1(c), $\beta_\pi = \mathbb{E}_{x_F}[\phi^K(Y_W)]$ and $\bar{\delta}_\pi = \mathbb{E}_{x_F}[\delta(Y_W)]$. (a) (Order flow is worthless.) Let $\varepsilon = 0$. For every rule π and every $x \in \mathcal{E}_W(0)$, $\beta_\pi(x) = \beta_\pi$, and π verifies with probability $\bar{\delta}_\pi$ under both causes. Let $\bar{\pi}$ be the rule that does not use Y_W : it verifies with probability $\bar{\delta}_\pi$; after verification, on $Z = z$, it contains with probability $\mathbb{E}_{x_F}[\delta \pi_z(Y_W)]/\bar{\delta}_\pi$; without verification it contains with probability $\mathbb{E}_{x_F}[(1 - \delta)\pi_0(Y_W)]/(1 - \bar{\delta}_\pi)$ (either ratio arbitrary in $[0, 1]$ when its denominator is 0). Then $\beta_{\bar{\pi}}(x) = \beta_\pi(x)$, $\alpha_{\bar{\pi}} = \alpha_\pi$ and $J(\bar{\pi}, x) = J(\pi, x)$ for every $x \in \mathcal{E}_W(0)$. (b) (The attacker takes the ceiling.) Let $\varepsilon = 0$. If $\beta_\pi < 1$, then $\text{BR}_0(\pi) = \{x_F + \Delta : \Delta \in \mathcal{B}_W, \|\Delta\|_2 \leq r, g^\top \Delta = \bar{\mathcal{F}}_W\}$, so every best response has excess harm $\bar{\mathcal{F}}_W$. If $\beta_\pi = 1$, then $\text{BR}_0(\pi) = \mathcal{E}_W(0)$ and $J(\pi, x)$ is the same for every $x \in \mathcal{E}_W(0)$. (c) (Equilibrium.) Let $\varepsilon = 0$, and let π_0^**

be the rule that ignores Y_W and applies Paper 2's two-threshold policy (Paper 2, Theorem 1) at the belief $q = p$, for the loss table with H_K replaced by $H_K + \bar{\mathcal{F}}_W$ (so with a replaced by a' ; write q'_1, q'_2 for its thresholds, $q'_1 = q'_2$ when its verification region is empty). Then (π_0^, x^*) , for any $x^* \in \text{BR}_0(\pi_0^*)$, is a strong Stackelberg equilibrium at $\varepsilon = 0$ in the class of all rules of Definition 5(iii), and in every class that contains π_0^* . Its cost J_0^* is Paper 2's optimal verify-first expected cost in the confounded case (Paper 2, Definition 4 and Corollary 1(ii)) for that loss table. (d) (All or nothing.) At $\varepsilon = 0$ the rule π_0^* verifies with probability $\mathbb{W}[q'_1 < p < q'_2]$ whatever Y_W , whereas π_x^{ver} with Paper 2's thresholds verifies with probability m_F under both causes, against every $x \in \mathcal{E}_W(0)$. If $\bar{\mathcal{F}}_W = 0$ (in particular at the state level with $H \geq n$ and (A, C) observable, Theorem 1(c)), then $q'_1 = q_1$ and $q'_2 = q_2$, and π_0^* verifies if and only if the prior lies in Paper 2's verification region. (e) (Continuity from above.) For $\varepsilon \in (0, 1]$, write $\bar{\beta}^* = \bar{\beta}_{\pi_0^*}$. Then $J(\pi_0^*, x) = J_0^* + p(1 - \bar{\beta}^*)(\mathcal{F}_W(\varepsilon) - \bar{\mathcal{F}}_W)$ for every $x \in \text{BR}_\varepsilon(\pi_0^*)$. Hence the strong Stackelberg cost at stealth level ε , in any class that contains π_0^* , is at most this value, and this bound tends to J_0^* as $\varepsilon \downarrow 0$.*

Proof: Supplementary Material, Section S1.8.

Part (d) contrasts two verification decisions. Paper 2's committed rule verifies whenever the nominal posterior falls in its verification interval, which under the fault law has probability m_F (Definition 6); the Stackelberg rule at $\varepsilon = 0$ ignores the order flow and verifies for every observation or for none, according to the prior alone. Part (e) proves a bound on the Stackelberg cost, not the convergence of the Stackelberg verification thresholds as ε tends to zero; the verification region at $\varepsilon > 0$ is computed in Section V.

V. NUMERICAL STUDY

Propositions 5 and 6 give the operator's optimal commitment in two regimes: among threshold rules when the stealth constraint binds and the capture loss equals the containment saving, and for every rule at $\varepsilon = 0$. This section computes the strong Stackelberg commitment of Definition 8 outside these regimes, on random linear-Gaussian instances, and tests the two claims the theory leaves open: that the optimal commitment chooses a statistic that sees the attacker's harmful directions, and that against a strategic attacker the optimal verification region is wider than the non-strategic interval (q_1, q_2) of Paper 2 [2] where fault posteriors rarely reach it.

A. Design

Instances have state dimension n of 2, 3 and 4, output dimension 1, 2 and 3 respectively, output-level and state-level injection, and windows of length 1 and n , with 200 instances per setting of dimension and injection level (1200 in all); at the output level the harm ceiling is then non-zero almost surely, and at the state level with a window at least as long as the state dimension it is zero almost surely

(Theorem 1), so the contrast between the two appears in every setting. For each instance, prior p and stealth level ε , the operator's commitment is computed by an outer minimization of its cost J (Definition 7) over the nominal model and the threshold (threshold rules) or the two thresholds (rules with verification), with the attacker's ε -stealthy best response computed in closed form (Proposition 3). The Stackelberg commitment is compared with the *naive commitment*, the threshold rule committed to the attacker's unconstrained best payload $x_F + r g / \|g\|_2$, and, for rules with verification, with Paper 2's two-threshold rule at that nominal model. A statistic *sees* the attacker's best payload when its seeing index, the absolute cosine between the statistic's direction and that payload in the M_W geometry, is positive. The instance laws, the grids of p and ε , the verification parameters and the decision criteria were fixed before the computation; each outcome is reported pooled over settings, with the per-setting outcomes beside it.

B. Optimal commitment and the statistic it sees

A *problem* is one instance at one prior p and one stealth level ε . The claim is tested on the problems where the naive commitment is exploited, that is, where the attacker's ε -stealthy best response to it is detected only at the false-alarm rate (Proposition 3); it fails in a problem when the Stackelberg commitment does not see that response or does not cost less than the naive commitment. A constant Stackelberg rule (never or always contain) counts as not seeing: it detects every payload at its false-alarm rate, which is the property that defines not seeing in Proposition 3. The outcome is *inconclusive*, pooled and in every setting: the cost difference between the Stackelberg and the naive commitment lay within the optimizer's certified tolerance in more than half of the problems where the naive commitment is exploited, which the rule fixed before the computation reads as undecided at this power. The decision rule was fixed before the computation; under its stricter reading, a tolerance strictly above the difference, the same data give *refutes*, pooled and in every setting. The two readings differ because most of these problems are exact ties, in which the difference and the tolerance are both zero; the outcome reported is the one under the rule as fixed beforehand. Pooled, the claim fails in a share 0.817 of the 23791 problems where the naive commitment is exploited (0.789 to 0.857 across settings), and the difference lies within the tolerance in a share 0.648 of them (0.544 to 0.723).

The ties have a single form. In most problems where the naive commitment is exploited, the Stackelberg-optimal threshold commitment does not use the observation: it never contains at the low priors (7982 of the 8493 such problems at $p = 0.01$), always contains at the highest (1371 of 1371 at $p = 0.5$), and costs exactly what the naive commitment costs. Excluding the constant rules, the failure share falls to 0.097 (0.074 to 0.114 across settings). In this design, then, the optimal commitment does not

need to see the harmful direction, because at most priors containing on the observation is not worth its false alarms whatever the statistic; the paper does not claim that the optimal commitment must see. Proposition 5(b) is stated for $c_A = a$ and this study uses $c_A = 0$, so the study neither tests nor contradicts it; the pattern is consistent with its clause that, where no minimizing cutoff exists, the infimum is approached as rules tend to never or always containing. Reading the attacker's best response over the payload budget alone, instead of over ε -stealthy payloads, gives the same counts.

C. Verification region against a strategic attacker

The outcome is that the claim is *weakened*, pooled and in every setting. The Stackelberg verification interval starts below Paper 2's q_1 in most problems where the fault verification mass m_F is low, but in fewer than the share predicted beforehand, and the change is at least as large where m_F is high, where the Stackelberg rule often verifies every posterior below its upper threshold ($q_1^* = 0$). Against a strategic attacker, verification is thus placed more widely in much of the design, but this does not track how rarely fault posteriors reach Paper 2's interval, which was the conjectured mechanism. The outcome is the same whether thresholds on the edge of the search box are kept, counted as not widening, or excluded. As stated before the computation, where the Stackelberg direction differs from the harm direction, q_1^* and q_1 are posterior thresholds of different nominal models; the scale-free description of the region, its fault-law mass under both rules, is reported beside. The sensitivity points of the verification parameters were not run. Pooled, $q_1^* < q_1$ in a share 0.684 of the 17825 low- m_F problems (0.588 to 0.795 across settings) and 0.926 of the 44530 high- m_F problems (0.862 to 0.958), with median $|q_1^* - q_1|$ of 0.062 and 0.083 in the two groups. In the low group the median fault-law mass of the verification region is 9.10×10^{-6} under the Stackelberg rule and 9.56×10^{-5} under Paper 2's rule at the naive nominal model.

VI. DISCUSSION

Proposition 7 (Bounds that hold for every strategy pair). *Assume Assumption 1 and Assumption 3. Let $\varepsilon \in [0, 1]$, let π be any rule of Definition 5 (randomized or not), and let σ be any Borel probability measure on $\mathcal{E}_W(\varepsilon)$ (a mixed attacker strategy; a pure strategy x_K is $\sigma = \delta_{x_K}$). Write $\beta_\pi(\sigma) = \int \beta_\pi(x) \sigma(dx)$ and $h(\sigma) = \int h(x - x_F) \sigma(dx)$ ($h(\sigma)$ is well defined; so is $\beta_\pi(\sigma)$ for a rule without verification and, under Assumption 6, for a rule with verification). (a) (Detection; Proposition 1.) If π has no verification, $\alpha_\pi - \varepsilon \leq \beta_\pi(\sigma) \leq \alpha_\pi + \varepsilon$. If π has verification and Assumption 6 holds, then, with ϕ^K as in Proposition 1(c), $\mathbb{E}_{x_F}[\phi^K(Y_W)] - \varepsilon \leq \beta_\pi(\sigma) \leq \mathbb{E}_{x_F}[\phi^K(Y_W)] + \varepsilon$. (b) (Verification margin; Proposition 4.) If $\pi = \pi_x^{\text{ver}}$ and Assumption 6 holds, $(s_v + t_v - 1)m_F - \varepsilon \leq \beta_\pi(\sigma) - \alpha_\pi \leq (s_v + t_v - 1)m_F + \varepsilon$. (c) (Harm; Proposition 2, Theorem 1.) $h(\sigma) \leq \mathcal{F}_W(\varepsilon)$, and $\mathcal{F}_W(\varepsilon) \downarrow \bar{\mathcal{F}}_W = r \|\Pi_{\mathcal{B}_W} g\|_2$ as $\varepsilon \downarrow 0$,*

with the values of Theorem 1(b)–(d). In particular (a)–(c) hold at every Nash equilibrium of the simultaneous-play game (Definition 8) and at every strong Stackelberg equilibrium, at stealth level ε .

Proof: Supplementary Material, Section S1.9.

a) Simultaneous play: The results are stated for an operator who commits. Proposition 7 shows which of them do not depend on commitment. The detection bounds of Propositions 1 and 4 and the harm bounds of Proposition 2 and Theorem 1 hold for every rule (for a rule with verification, under Assumption 6) and every attacker strategy, pure or mixed, within total variation ε of the fault’s law. They therefore hold at every Nash equilibrium of the simultaneous-play game, as well as at the strong Stackelberg equilibrium. Two things depend on commitment. The first is the exploitation described by Proposition 3: the proposition remains true for every fixed threshold rule, but in a Nash equilibrium the operator’s rule is a best response to the attacker’s actual strategy, not a rule committed to a nominal attack model, so the blind directions it exhibits need not be available to the attacker. The second is the optimal commitment: Proposition 5(b), Proposition 6(c) and (e), and the numerical study of Section V are statements about the strong Stackelberg equilibrium. Simultaneous-play formulations of detection against stealthy attackers include the Bayesian Nash equilibrium of [15] and the zero-sum saddle point of [18].

b) Limits of validity: (i) *Separate channel.* The verification margin requires that verification read a channel the attacker cannot influence (Assumption 6). (ii) *Payoff.* The harm ceiling is stated for excess harm, the harm beyond what the imitated fault causes. Under an absolute-harm payoff the ceiling is the fault’s own harm plus the excess-harm ceiling (Theorem 1(e)); where the latter is zero, the attacker can do no more than the fault it imitates. (iii) *Stealth level.* The ceiling is a limit as ε tends to zero at a fixed payload budget. At a fixed $\varepsilon > 0$ the frontier can still be large along weakly observable directions. (iv) *Information.* The game has complete information (Assumption 2): the attacker knows the fault instance it imitates, and the operator knows the attacker’s harm vector, budget, capture loss and stealth level. The bounds of Proposition 7 use neither, so they hold without this assumption; the optimal commitment does not. An attacker uncertain about the fault law, or an operator uncertain about the attacker’s parameters, is left to future work. (v) *Model.* The results hold in the linear-Gaussian model of Assumption 1, with one fixed fault instance and a finite window. Zero excess harm at the state level needs a window at least as long as the state dimension and an observable pair (A, C) , which holds for almost every A . (vi) *Numerical study.* Section V uses $c_A = 0$ and one cost triple, so the constant-rule pattern it reports is shown for that payoff only.

VII. CONCLUSION

Against an operator who commits to its rule, an attacker who must stay within total variation ε of a fault’s obser-

vation law faces a stealth–harm frontier. The excess harm it can extract is nondecreasing and continuous in ε , and strictly below its unconstrained value below a computable stealth level unless its harm direction is invisible on the window (Proposition 2). As ε tends to zero it falls to the harm along the blind subspace (Lemma 1, Theorem 1): a closed form at the output level, and zero excess harm under state-level injection for almost every system matrix. No rule without verification contains such an attacker with probability more than ε above its false-alarm rate (Proposition 1). A threshold rule committed to a nominal attack model contains payloads its statistic does not see only at its false-alarm rate (Proposition 3). A two-threshold rule with verification on a separate channel adds a detection margin that no order-flow choice can erase, in proportion to how often fault posteriors reach the verification interval (Proposition 4). The detection and harm bounds hold for every strategy pair, and hence also under simultaneous play (Proposition 7). We have given two further results. When the harm direction has no component in the blind subspace and the stealth constraint binds, the rule to commit to instead of a nominal one plans against the frontier’s maximizer M_W^+g ; among threshold rules with the same false-alarm rate it minimizes the attacker’s value and is the only one that leaves no harmful payload unseen, and when the capture loss equals the operator’s containment saving a Stackelberg-optimal commitment among threshold rules can be taken of this form where a minimizing cutoff exists (Proposition 5); the naive commitment to the harm direction has this property only when that direction is an eigenvector of the stealth matrix. In the limiting case of exact imitation the game reduces to Paper 2’s confounded case, with the leave-a-compromise cost raised by the harm ceiling: the operator verifies at the prior or not at all, and its cost bounds the Stackelberg cost from above for every ε , up to the excess of the frontier over the ceiling (Proposition 6). They rest on a separate verification channel, an excess-harm payoff, complete information and a linear-Gaussian model (Section VI). Outside these two regimes, a numerical study, not a proof, finds that in random instances the optimal threshold commitment usually ignores the observation, and that the strategic verification region widens but not preferentially where fault posteriors rarely reach Paper 2’s interval (Section V).

REFERENCES

- [1] C. Mejrissi, “Attack–fault confounding in a linear model of the electronic trading gateway is generically fragile under state-level injection,” 2026, working paper. Manuscript and code: <https://github.com/mejrissi/Thesis>, directory papers/paper1.
- [2] —, “Verification as a costed action: When checking beats containing under attack–fault ambiguity,” 2026, working paper.
- [3] —, “Graded containment with an observability price: Sequential response to ambiguous trading incidents,” 2026, in preparation.
- [4] D. Korzhyk, Z. Yin, C. Kiekintveld, V. Conitzer, and M. Tambe, “Stackelberg vs. Nash in security games: An extended investigation of interchangeability, equivalence, and uniqueness,” *Journal of Artificial Intelligence Research*, vol. 41, pp. 297–327, 2011.

- [5] M. Brückner and T. Scheffer, “Stackelberg games for adversarial prediction problems,” in *Proceedings of the 17th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2011, pp. 547–555.
- [6] M. Hardt, N. Megiddo, C. Papadimitriou, and M. Wootters, “Strategic classification,” in *Proceedings of the 2016 ACM Conference on Innovations in Theoretical Computer Science*, 2016, pp. 111–122.
- [7] F. Pasqualetti, F. Dörfler, and F. Bullo, “Attack detection and identification in cyber-physical systems,” *IEEE Transactions on Automatic Control*, vol. 58, no. 11, pp. 2715–2729, 2013.
- [8] C.-Z. Bai, F. Pasqualetti, and V. Gupta, “Data-injection attacks in stochastic control systems: Detectability and performance tradeoffs,” *Automatica*, vol. 82, pp. 251–260, 2017.
- [9] X.-X. Ren and G.-H. Yang, “Kullback–Leibler divergence-based optimal stealthy sensor attack against networked linear quadratic Gaussian systems,” *IEEE Transactions on Cybernetics*, vol. 52, no. 11, pp. 11 539–11 548, 2022.
- [10] J. Shang, H. Yu, and T. Chen, “Worst-case stealthy innovation-based linear attacks on remote state estimation under Kullback–Leibler divergence,” *IEEE Transactions on Automatic Control*, vol. 67, no. 11, pp. 6082–6089, 2022.
- [11] L.-W. Mao and G.-H. Yang, “Optimal stealthy attack against remote state estimation with parameterized ϵ -stealthiness,” *IEEE Transactions on Industrial Informatics*, vol. 22, no. 2, pp. 1598–1606, 2026.
- [12] P. Griffioen, B. H. Krogh, and B. Sinopoli, “Ensuring resilience against stealthy attacks on cyber-physical systems,” *IEEE Transactions on Automatic Control*, vol. 69, no. 12, pp. 8234–8246, 2024.
- [13] A. T. Nguyen, A. M. H. Teixeira, and A. Medvedev, “Security allocation in networked control systems under stealthy attacks,” *IEEE Transactions on Control of Network Systems*, vol. 12, no. 1, pp. 216–227, 2025.
- [14] A. T. Nguyen, S. C. Anand, and A. M. H. Teixeira, “Scalable and optimal security allocation in networks against stealthy injection attacks,” *IEEE Transactions on Automatic Control*, vol. 71, no. 5, pp. 3455–3462, 2026.
- [15] D. Umsonst, S. Saritaş, G. Dán, and H. Sandberg, “A Bayesian Nash equilibrium-based moving target defense against stealthy sensor attacks,” *IEEE Transactions on Automatic Control*, vol. 69, no. 3, pp. 1659–1674, 2024.
- [16] N. Dalvi, P. Domingos, Mausam, S. Sanghai, and D. Verma, “Adversarial classification,” in *Proceedings of the Tenth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2004, pp. 99–108.
- [17] Y. Hu, J. Chen, and Q. Zhu, “Game-theoretic Neyman–Pearson detection to combat strategic evasion,” *IEEE Transactions on Information Forensics and Security*, vol. 20, pp. 516–530, 2025.
- [18] T. Tanaka, K. Sawada, Y. Watanabe, and M. Iwamoto, “Covert vehicle misguidance and its detection: A hypothesis testing game over continuous-time dynamics,” *IEEE Control Systems Letters*, vol. 8, pp. 2889–2894, 2024.
- [19] K. Zhang, C. Keliris, T. Parisini, and M. M. Polycarpou, “Identification of sensor replay attacks and physical faults for cyber-physical systems,” *IEEE Control Systems Letters*, vol. 6, pp. 1178–1183, 2022.
- [20] A. Naha, A. M. H. Teixeira, A. Ahlén, and S. Dey, “Quickest detection of deception attacks on cyber-physical systems with a parsimonious watermarking policy,” *Automatica*, vol. 155, p. 111147, 2023.
- [21] V. P. Crawford and J. Sobel, “Strategic information transmission,” *Econometrica*, vol. 50, no. 6, pp. 1431–1451, 1982.
- [22] I.-K. Cho and D. M. Kreps, “Signaling games and stable equilibria,” *The Quarterly Journal of Economics*, vol. 102, no. 2, pp. 179–221, 1987.
- [23] E. Kamenica and M. Gentzkow, “Bayesian persuasion,” *American Economic Review*, vol. 101, no. 6, pp. 2590–2615, 2011.
- [24] X. Tao, A. Day, L. Ling, and S. Drapeau, “On detecting spoofing strategies in high-frequency trading,” *Quantitative Finance*, vol. 22, no. 8, pp. 1405–1425, 2022.